

Software Conceptual Design

Next-Generation Ground-based Solar Observing Network

ngGONG Software Team

2026-11-06

Table of contents

I Design 4

1 Scope and Document Relationships 5

2 Architectural Drivers and Constraints 6

3 Architectural Overview 7

4 Major HLS Software Components 8

4.1 OCS . . . 8
4.2 SCS . . . 8
4.3 TuCS . . . 8
4.4 TeCS . . . 8
4.5 GLI and Other Instruments . . . 8
4.6 Safety Interlock System . . . 9

5 Major DMS Software Components and Data Lifecycle 10

6 Selected Sequence Diagrams and Conceptual Behavior 11

7 Deployment and Communications Concepts 12

8 Major Internal and External Interfaces 13

9 Information Assurance and Resilience Concepts 14

10 Major Design Decisions and Feasibility Evidence 15

11 Technical Risks and Planned Mitigations 16

12 Development Approach and Path to PDR 17

II Requirements Allocation 18

13 OCS Requirements Allocation 19

14 SCS Requirements Allocation 20

15 DMS Requirements Allocation 21

16 SIS Requirements Allocation 22

III Architecture Design Summary **23**

17 Architectural Design Summary **24**

Part I

Design

1 Scope and Document Relationships

> **Draft stub:** This section will define the software system boundaries and responsibilities covered by this document.

Planned content includes:

- the scope of High Level Software and Controls (HLS & C) and the Data Management System (DMS), both individually and as an integrated system;
- the boundaries between HLS & C, DMS, other ngGONG systems, and external systems; and
- the relationships between this document and the applicable Design Requirements Documents (DRDs), operational concept documentation, system architecture, and supporting plans.

This section will identify authoritative source documents rather than repeat their requirements, operational narratives, or detailed processes.

2 Architectural Drivers and Constraints

> **Draft stub:** This section will identify the requirements, assumptions, and constraints that materially influence the conceptual software design.

Planned content includes:

- requirements that drive major architectural choices;
- significant performance, capacity, availability, interoperability, information assurance, and lifecycle constraints;
- external dependencies and constraints imposed by other ngGONG systems; and
- assumptions, open questions, and unresolved allocations that affect the design.

Requirements will be referenced by their authoritative identifiers and will not be duplicated here.

3 Architectural Overview

> **Draft stub:** This section will present the integrated conceptual architecture for HLS & C and DMS.

Planned content includes:

- the major software systems, subsystems, and shared services;
- their principal responsibilities and ownership boundaries;
- the major control, science-data, metadata, status, and engineering-data paths; and
- the relationships among station, observatory, and data-management elements.

This section will include or reference the system-context and integrated architecture views needed to explain the design.

4 Major HLS Software Components

> **Draft stub:** This section will describe the responsibilities, decomposition, and relationships of the major HLS & C components at a conceptual level.

The descriptions will identify major responsibilities, interfaces, control and data paths, dependencies, and unresolved design questions without specifying implementation-level APIs, message definitions, or state machines.

4.1 OCS

TBD: Describe the conceptual responsibilities and decomposition of the Observatory Control System (OCS).

4.2 SCS

TBD: Describe the conceptual responsibilities and decomposition of the Station Control System (SCS).

4.3 TuCS

TBD: Describe the conceptual responsibilities and decomposition of TuCS.

4.4 TeCS

TBD: Describe the conceptual responsibilities and decomposition of TeCS.

4.5 GLI and Other Instruments

TBD: Describe the conceptual responsibilities and software relationships of GLI and other instruments.

4.6 Safety Interlock System

TBD: Describe the conceptual architecture of the Safety Interlock System (SIS), including allocated safety functions, safety authority, independence, major components, protected equipment interfaces, and fault-response concepts.

This subsection will reference the SIS DRD, hazard analysis, and cause-and-effect matrix rather than duplicate them.

5 Major DMS Software Components and Data Lifecycle

> **Draft stub:** This section will describe the responsibilities and conceptual decomposition of DMS and the end-to-end lifecycle of ngGONG data.

Planned content includes:

- data ingest and validation;
- processing allocated to DMS;
- metadata and provenance management;
- storage and preservation;
- data discovery and access; and
- principal exchanges with HLS & C and external consumers.

This section will identify ownership boundaries, major data states, and unresolved allocations without defining detailed schemas or database designs.

6 Selected Sequence Diagrams and Conceptual Behavior

> **Draft stub:** This section will show how major software components cooperate in representative operational and failure scenarios.

Candidate scenarios include:

- an observing sequence through data ingest and availability;
- fault handling and recovery; and
- disconnected station operation and reconciliation.

The selected views will explain software cooperation and significant conceptual state behavior. They will reference, rather than repeat, the operational narratives in the applicable operational concept documentation.

7 Deployment and Communications Concepts

> **Draft stub:** This section will describe the conceptual deployment and communications architecture.

Planned content includes:

- placement of major software components at stations and observatory or data-management facilities;
- infrastructure and platform dependencies;
- major network paths and communications patterns;
- trust, availability, and failure boundaries; and
- operation during degraded or disconnected conditions.

This section will not define deployment scripts, detailed network configurations, or implementation-level protocol specifications.

8 Major Internal and External Interfaces

> **Draft stub:** This section will identify the significant interfaces within the software architecture and between ngGONG software and external systems.

For each major interface, planned content includes:

- ownership and responsibility boundaries;
- principal commands, data, metadata, status, or engineering-information exchanges;
- conceptual communication semantics and interoperability constraints; and
- authoritative interface definitions or Interface Control Documents (ICDs), where available.

Particular attention will be given to interfaces between HLS & C and DMS. This section will not include complete ICDs, detailed APIs, Data Distribution Service (DDS) topic definitions, or Protocol Buffer definitions.

9 Information Assurance and Resilience Concepts

> **Draft stub:** This section will summarize the information assurance and resilience implications of the conceptual architecture.

Planned content includes:

- trust boundaries and access concepts;
- protection of software, data, metadata, and control paths;
- availability, fault containment, backup, and recovery concepts;
- resilience during network or service disruptions; and
- applicable constraints, dependencies, assumptions, and open questions.

Detailed cybersecurity configurations, procedures, and implementation controls will remain in their applicable supporting artifacts.

10 Major Design Decisions and Feasibility Evidence

> **Draft stub:** This section will summarize the major architectural decisions and the evidence supporting the technical feasibility of the conceptual design.

Planned content includes:

- alternatives considered, selection criteria, and decision rationale;
- architectural consequences and unresolved issues;
- performance and capacity estimates, including applicable margins;
- technology maturity; and
- relevant analyses, simulations, prototypes, or other supporting evidence.

Detailed trade studies and any separate Architecture Decision Summary will be referenced rather than duplicated. Demonstrated capabilities will be clearly distinguished from estimates, assumptions, and planned work.

11 Technical Risks and Planned Mitigations

> **Draft stub:** This section will identify the principal technical risks that affect the conceptual software design and the work needed to reduce them.

Planned content includes:

- the architectural area or uncertainty associated with each risk;
- the potential design consequence;
- planned mitigation or risk-reduction activity;
- required analysis, prototype, or design evidence; and
- references to the authoritative project risk register.

This section will summarize design implications and will not replace the project risk-management process or controlled risk records.

12 Development Approach and Path to PDR

> **Draft stub:** This section will identify the design work needed to advance the software architecture from the Conceptual Design Review (CoDR) to the Preliminary Design Review (PDR).

Planned content includes:

- conceptual integration and verification approaches;
- major technical dependencies and unresolved design work;
- planned analyses, prototypes, and other risk-reduction activities;
- required requirements allocation and design-compliance evidence; and
- technical inputs needed by cost, acquisition, operations, lifecycle, and disposition planning.

Detailed development processes, verification procedures, schedules, cost estimates, acquisition plans, and operations or disposition plans will remain in their respective documents.

Part II

Requirements Allocation

13 OCS Requirements Allocation

> **Draft stub:** This section will summarize requirements allocation and design compliance for the Observatory Control System (OCS).

Planned content includes:

- references to the authoritative science and system requirements hierarchy and applicable High Level Software (HLS) Design Requirements Document (DRD);
- mapping of applicable requirement identifiers to major OCS design elements, consistent with the project's Spicy SE traceability;
- allocation of shared-interface responsibilities and applicable interoperability requirements;
- supporting design evidence and proposed verification approaches; and
- identified non-compliances, unresolved allocations, and planned resolutions.

This section will reference controlled allocation and compliance records rather than duplicate the requirements catalog. Assumptions and proposed allocations will be distinguished from established requirements.

TBD: Identify the authoritative records, applicable requirements, design mappings, and compliance status.

14 SCS Requirements Allocation

> **Draft stub:** This section will summarize requirements allocation and design compliance for the Station Control System (SCS).

Planned content includes:

- references to the authoritative science and system requirements hierarchy and applicable HLS DRD;
- mapping of applicable requirement identifiers to major SCS design elements, consistent with the project's Spicy SE traceability;
- allocation of responsibilities at station control, instrument, safety, and data-management interfaces, including applicable interoperability requirements;
- supporting design evidence and proposed verification approaches; and
- identified non-compliances, unresolved allocations, and planned resolutions.

This section will reference controlled allocation and compliance records rather than duplicate the requirements catalog. Assumptions and proposed allocations will be distinguished from established requirements.

TBD: Identify the authoritative records, applicable requirements, design mappings, and compliance status.

15 DMS Requirements Allocation

> **Draft stub:** This section will summarize requirements allocation and design compliance for the Data Management System (DMS).

Planned content includes:

- references to the authoritative science and system requirements hierarchy and DMS DRD;
- mapping of applicable requirement identifiers to major DMS design elements, consistent with the project's Spicy SE traceability;
- coverage of ingest, processing where allocated, metadata, storage, preservation, discovery, and access;
- allocation of shared-interface responsibilities and applicable interoperability requirements, particularly at the HLS and Controls boundary;
- supporting design evidence and proposed verification approaches; and
- identified non-compliances, unresolved allocations, and planned resolutions.

This section will reference controlled allocation and compliance records rather than duplicate the requirements catalog. Assumptions and proposed allocations will be distinguished from established requirements.

TBD: Identify the authoritative records, applicable requirements, design mappings, and compliance status.

16 SIS Requirements Allocation

> **Draft stub:** This section will summarize requirements allocation and design compliance for the Safety Interlock System (SIS).

Planned content includes:

- references to the authoritative science and system requirements hierarchy, SIS DRD, hazard analysis, and cause-and-effect matrix;
- mapping of applicable requirement identifiers and allocated safety functions to major SIS design elements, consistent with the project's Spicy SE traceability;
- supporting design evidence for safety authority, independence, fault response, and interfaces with HLS and protected equipment;
- allocation of shared-interface responsibilities and applicable interoperability requirements;
- proposed verification approaches; and
- identified non-compliances, unresolved allocations, and planned resolutions.

This section will reference controlled allocation and compliance records rather than duplicate the requirements catalog or safety analyses. Assumptions and proposed allocations will be distinguished from established requirements.

TBD: Identify the authoritative records, applicable requirements, design mappings, and compliance status.

Part III

Architecture Design Summary

17 Architectural Design Summary

> **Draft stub:** This section will provide concise summaries of key architectural trade studies and decisions for HLS and Controls and DMS.

Each summary is expected to occupy approximately one or two pages and identify:

- the architectural question and alternatives considered;
- the selection criteria and supporting rationale;
- the decision status, including whether a choice is proposed or established;
- the consequences, assumptions, and unresolved issues; and
- references to detailed trade studies, analyses, and supporting design evidence.

The summaries will complement the Major Design Decisions and Feasibility Evidence section and reference detailed material rather than duplicate it.

TBD: Identify the decisions and trade studies to summarize and add references to their authoritative records.